Simple Harmonic Motion Pendulum

Arduino Physics Apparatus

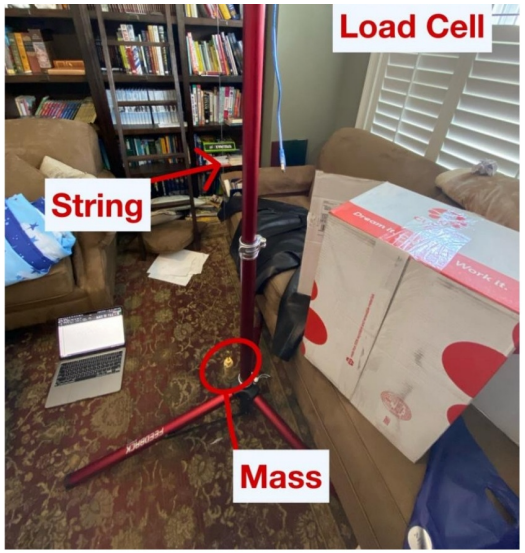
Designed and documented a two-sensor pendulum experiment to validate simple harmonic motion. The apparatus used a load cell to record changing vertical tension and a sonic ranger to track position over time, then used Excel analysis to identify sinusoidal behavior, damping, period, amplitude, and sensor error.

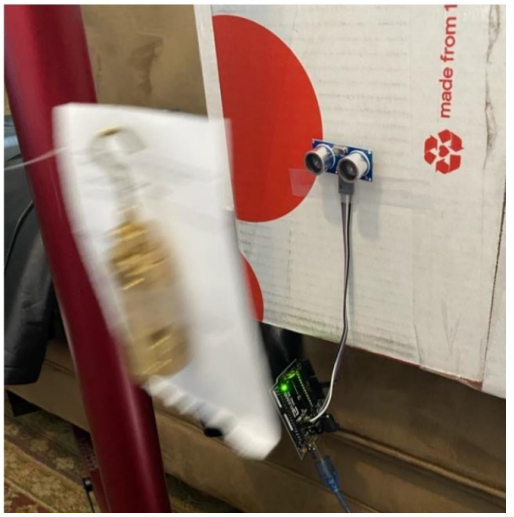
PHYSICS SHM + damping

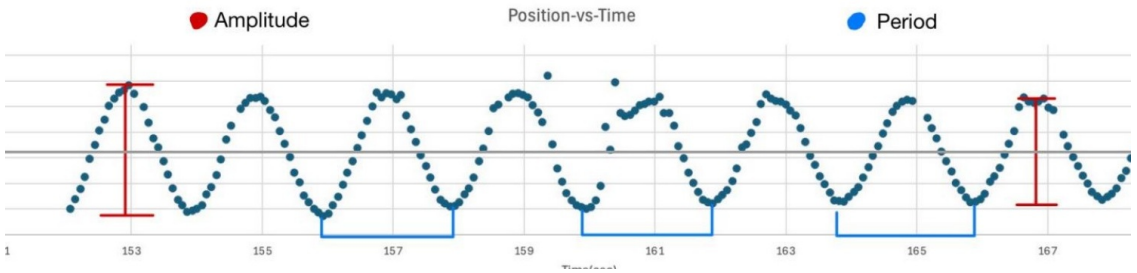
HARDWARE Arduino, HX711, load cell, sonic ranger

ANALYSIS CSV data + Excel plots

OUTPUT 11-page lab + repeatable procedure







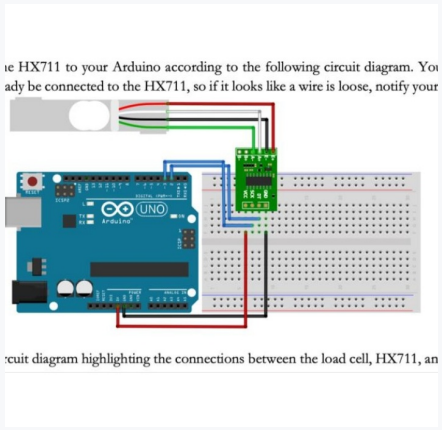
Research-style build: instrument, measure, analyze

PART 1

Changing Tension Force

A hanging mass was suspended from a load cell connected to an HX711 amplifier. As the pendulum swung, the changing vertical component of tension produced a sinusoidal force signal over time.





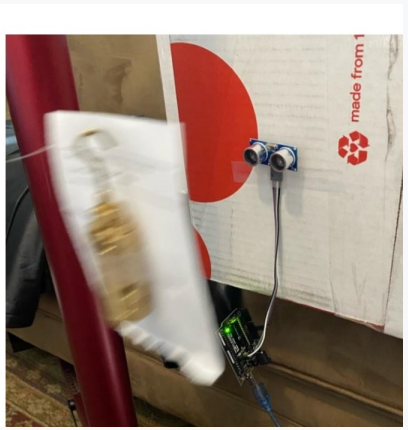
IMPLEMENTATION NOTES

Calibrated raw HX711 values using a known hanging mass; converted serial readings into apparent mass and force; exported time-series data for graphing and interpretation.

PART 2

Position Tracking

A sonic ranger measured pendulum position as a function of time. A flat paper target attached to the swinging mass improved sensor reflection and produced a clearer oscillatory position signal.

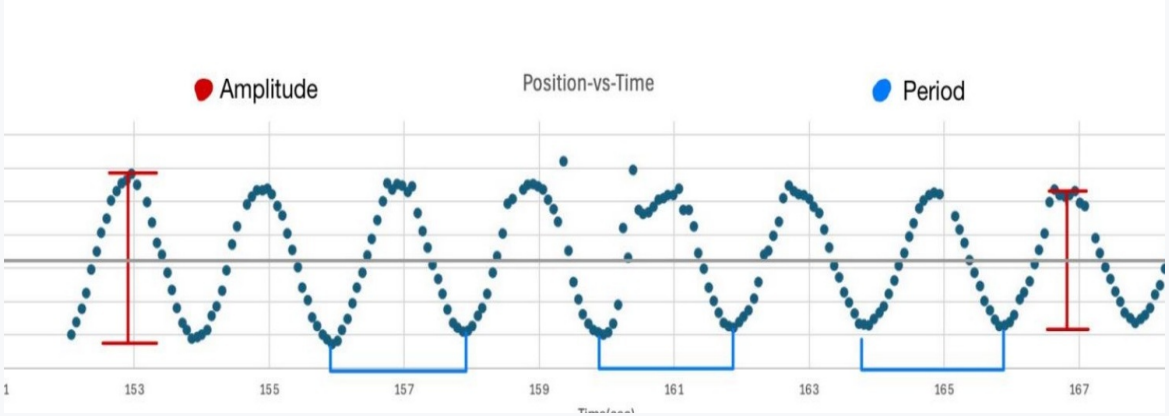




VALIDATION LOGIC

The force and position data were expected to follow sinusoidal patterns. Consistent period with decaying amplitude supported the SHM model with real-world damping.

OBSERVED WAVEFORM



What the data showed

- Position and force data produced sinusoidal curves.
- Amplitude decreased over time due to damping.
- Period remained approximately constant for small oscillations.
- Spurious sonic-ranger spikes were identified as sensor error.

Portfolio Summary

This project shows the full experimental workflow: building a sensor-based apparatus, calibrating a load cell, collecting serial data, graphing time-series measurements, and interpreting real-world deviations from ideal simple harmonic motion. It demonstrates mechanical setup, Arduino programming, instrumentation, troubleshooting, and physics-based analysis.

INSTRUMENTATION

Load cell, HX711, sonic ranger

SOFTWARE

Arduino IDE, serial monitor, Excel

PHYSICS

SHM, period, amplitude, damping

ANALYSIS

Filtering invalid readings and interpreting trends

Janssen Effect Granular Materials Rig

LOW-COST EXPERIMENTAL APPARATUS | LOAD CELL + ARDUINO + GRANULAR PHYSICS



Suspended load-cell apparatus with isolated cardboard tube and bottom plate.



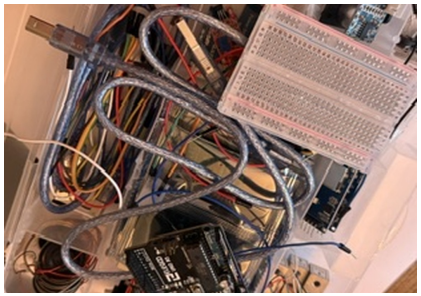
Rice measured in controlled increments.



Granular column during testing.



Suspended bottom plate/load path.



Arduino/HX711 electronics setup.

Research Question

Does the full weight of a granular column reach the bottom plate, or is force redirected into the tube walls through grain-wall friction?

Core Idea

Rice forms internal force chains. As the column grows taller, more load is carried sideways into the walls, so the bottom plate feels only a fraction of the total weight.

275 g
total rice added
highest recorded trial point

80.7 g
equivalent bottom reading
force reaching load plate

29.3%
weight reaching bottom
about 70.7% redirected

50x
averaged sensor samples
per button-triggered reading

Prototype note: early trials revealed a force-path error when the tube contacted the load plate; the final setup isolated the tube before recalibration.

Method, Data Pipeline, and Early Results

Experimental Method

1. **Isolate tube from plate**
   The cardboard tube is supported separately so wall-carried force does not enter the load cell.

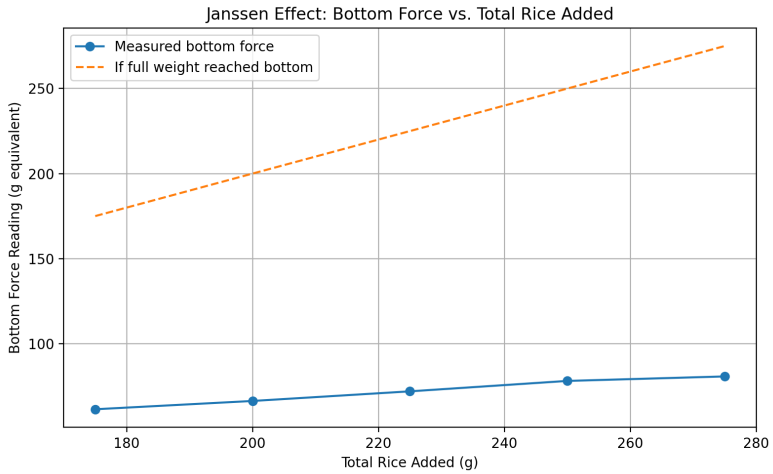
2. **Calibrate load sensor**
   HX711 raw readings are converted to mass/force using a 108 g calibration block.

3. **Add rice in increments**
   Rice is added in 25 g steps; the system settles before each reading.

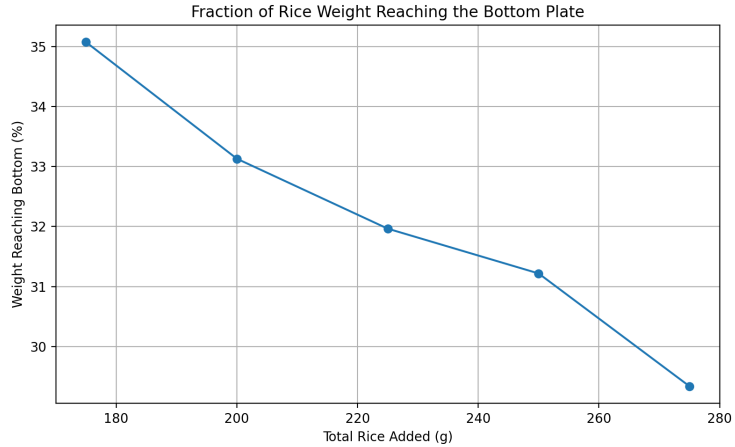
4. **Record averaged force**
   A button triggers a 50-sample average and prints one clean CSV row.

```
waitForButtonPress();
measurementNumber++;
long RDG = averageRawReading(50);
float mass_kg = knownMass_kg *
  (RDG - RDG_tare) /
  (RDG_known - RDG_tare);
float force_N = mass_kg * 9.80665;
Serial.println(force_N, 6);
```

Button-triggered data collection avoids noisy continuous sampling.



Measured bottom force grows much slower than total material weight.



The fraction of total weight reaching the bottom decreases as the column height increases.

Preliminary conclusion

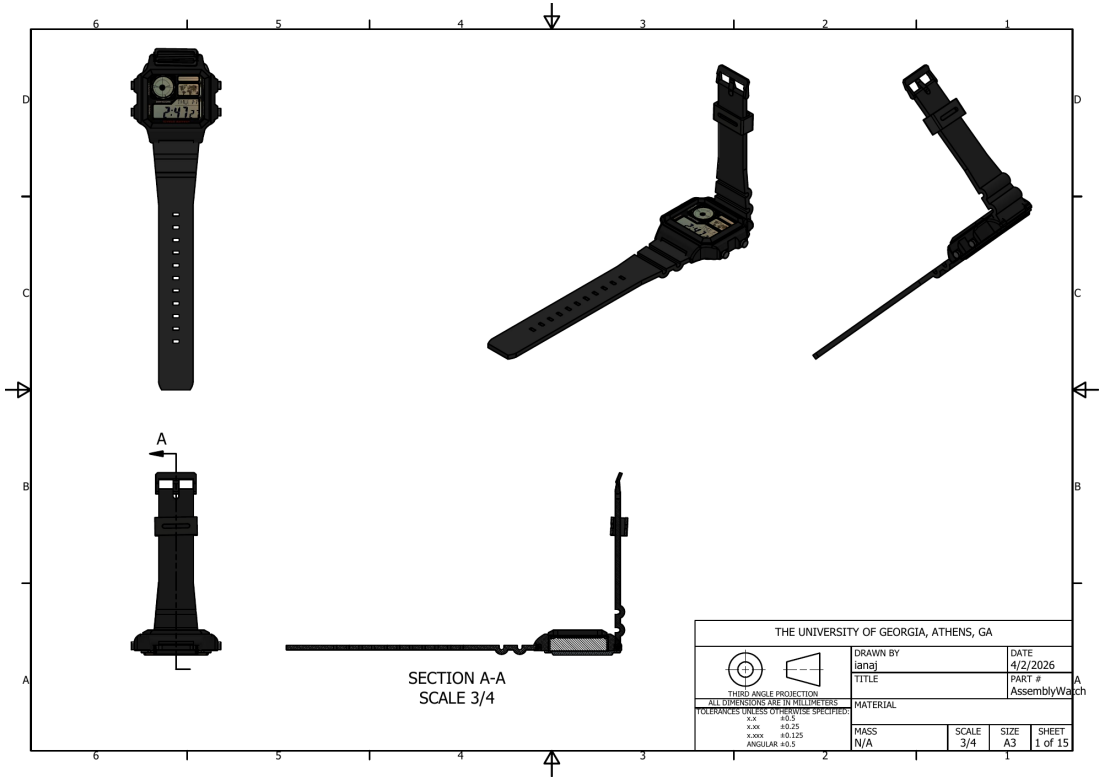
Adding rice increased total column weight much faster than the load measured at the bottom plate, consistent with Janssen-style force redirection into the tube walls.

Reverse Engineering a Digital Watch

CAD dissection, part modeling, assembly, and technical documentation

CAD CASE STUDY

Casio-style wristwatch



Full assembly drawing with orthographic, section, and isometric views.

Project Goal

Disassemble a consumer wristwatch, identify its components, model the parts in CAD, and rebuild the device as a complete digital assembly.

Scope

Modeled the main body, face plate, main module, back plate, long/short bands, band keeper, and two spring bars as an 8-item assembly.

Deliverables

Produced a bill of materials, component documentation, dimensioned part drawings, and an exploded assembly view.



Physical dissection photos and component layout.

Process: From Object to CAD Assembly

1. Disassemble

Separated the watch into functional components and documented the assembled and disassembled states with reference photos.

2. Measure

Created working sketches and dimensional references before building CAD geometry.

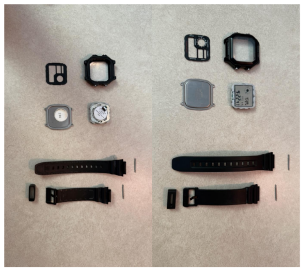
3. Model

Recreated individual parts with attention to shape, proportions, holes, fillets, reliefs, and material assignments.

4. Assemble

Combined parts into a full assembly and produced drawing sheets using standard engineering conventions.

Component Dissection



COMPONENTS

Bill of Materials

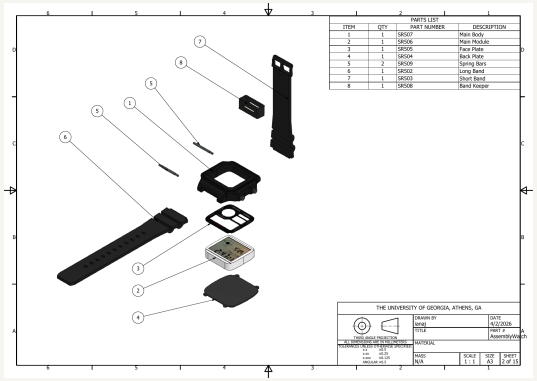


BILL OF MATERIALS

PARTS LIST

ITEM	PART NUMBER	QUANTITY	DESCRIPTION	MATERIAL
1	SR503	1	Long Band	Rubber
2	SR502	1	Short Band	Rubber
3	SR504	1	Back Plate	Steel
4	SR505	1	Face Plate	Plastic
5	SR506	1	Main Module	Plastic
6	SR507	1	Main Body	Plastic
7	SR508	1	Band Keeper	Rubber
8	SR509	2	Spring Bars	Aluminum

Exploded Assembly

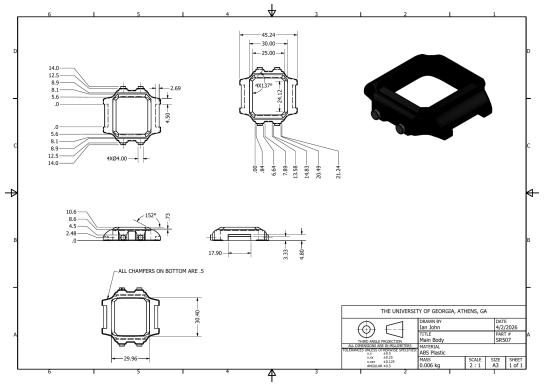


What this project demonstrates

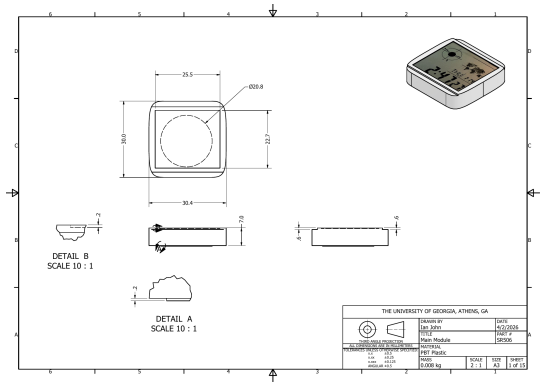
The project shows a complete reverse-engineering workflow: observe a real object, break it into manufacturable parts, translate physical geometry into CAD features, and communicate the design through professional drawing sheets. The strongest part of the work is the transition from a small, irregular consumer product into a clean assembly with consistent part names, materials, dimensions, and exploded-view documentation.

Technical Documentation Highlights

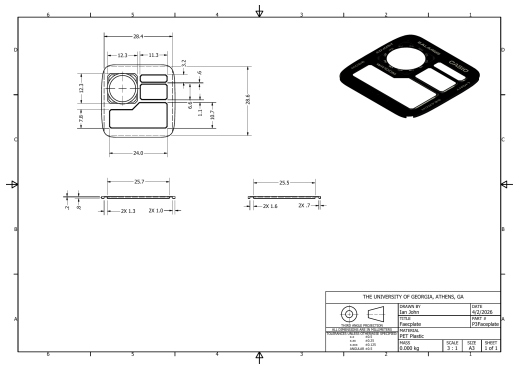
Main Body



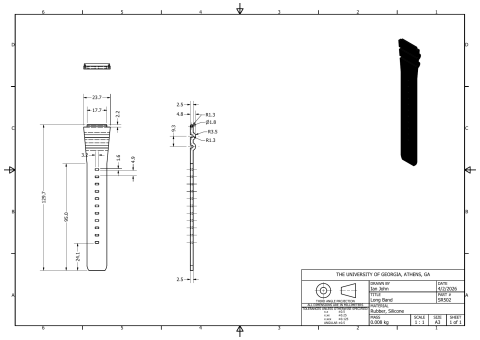
Main Module



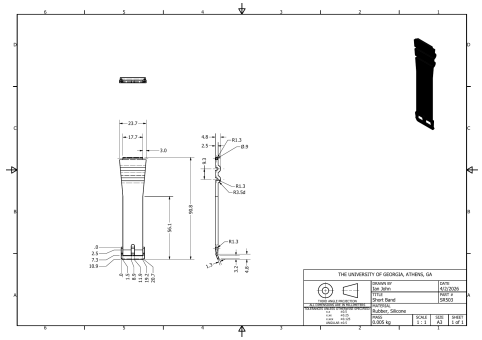
Face Plate



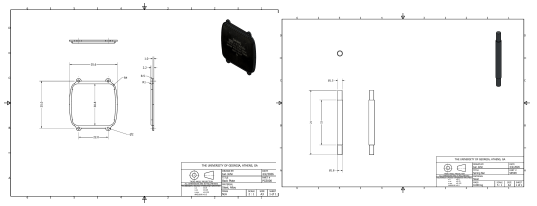
Long Band



Short Band



Back Plate + Spring Bar



Solid modeling | Reverse engineering | Assembly design | Dimensioned drawings | BOM creation | Third-angle projection

LLM Self-Correction Experiment

Testing whether one language model can improve another model's accuracy through iterative critique

AI EXPERIMENT | PROMPTING

Project Overview

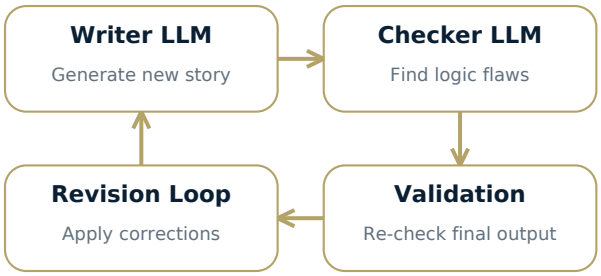
I designed a two-model LLM experiment around Alice Walker's short story Roselily. One model acted as a writer, generating a new perspective while preserving the original story logic. A second model acted as a checker, auditing the draft for continuity errors, point-of-view violations, and unsupported inferences.

The experiment was not about whether the prose sounded good. The central test was whether an AI-generated literary transformation could become more accurate when a separate AI repeatedly compared it against the source constraints.

Research Question

Can a checker LLM improve the factual and logical reliability of another LLM's creative output by repeatedly identifying errors and forcing revision?

Experimental System



What the Checker Evaluated

- Point-of-view limits: what child narrators could plausibly know
- Physical continuity: staging, space, and observable actions
- Chronology: keeping the original event sequence intact
- Semantic drift: preventing repeated paraphrase from altering meaning
- Unsupported inference: separating observed evidence from model invention

2	5+	4	0
LLM roles	error classes detected	major revision rounds	final logic errors flagged

Iteration Results: Accuracy Through Revision

Core Finding

The first draft sounded convincing, but the checker LLM found that fluency hid several accuracy problems. Most errors were not grammar mistakes; they were logic and perspective errors where the generated narration knew too much, borrowed adult symbolism, or slightly violated the original staging.

What the experiment shows

A second LLM can improve output reliability when it is assigned a narrow verification role, given the source constraints, and forced to explain specific errors rather than simply approve or rewrite.

Error Classes Caught by the Checker

Checker found	Why it mattered	Revision made
Porch rail staging	Children standing on a narrow rail was physically implausible.	Moved them to the steps by the rail.
Harvard detail	The child narrators should not know an adult-specific proper noun.	Changed to "some big college."
Religious certainty	The source makes the groom's religion partly unreadable to the community.	Kept it as visible behavior and rumor.
Adult metaphor	"Stalk without roots" belonged to Roselily's interiority.	Replaced with child-level physical imagery.
Tense/quotation drift	Small phrasing choices implied direct knowledge or broke present tense.	Removed quote framing and restored present-tense narration.

Revision Loop



Interpretation: The experiment did not show that LLMs are automatically reliable. It showed the opposite first: the writer model made plausible-looking mistakes. Reliability improved only after the checker model was constrained to audit logic against the source and the output was revised through multiple rounds.

Outcome, Evidence, and Technical Takeaways

Final Output Standard

The final version was judged by the checker system as perspective-tight: child narrators remained limited to what they could see, hear, or reasonably repeat as family talk. The experiment's success criterion was not literary beauty alone, but logical consistency after adversarial review.

Initial problem pattern

The writer model produced elegant prose that overreached: it introduced adult-level knowledge, implied direct access to private thoughts, and created small physical staging issues.

Final result pattern

The final output preserved story logic by anchoring narration in observable details, concrete language, and clearly attributed family talk rather than hidden motives or unsupported inference.

Technical Takeaways

Role separation

Writer and checker prompts created a useful generation-verification loop.

Constraint-based evaluation

The checker worked best when asked to audit specific categories of error.

Iteration improved accuracy

The first output was inaccurate in subtle ways; revision reduced those failures.

Semantic drift risk

Repeated paraphrase can improve one issue while slowly changing meaning elsewhere.

Self-checking has limits

The models were not fully independent, so final claims require cautious wording.

Validation mindset

The project treated AI output as something to test, not something to trust automatically.

Skills Demonstrated

Prompt engineering | LLM evaluation | iterative testing | logical validation | failure analysis | source-constrained generation | AI reliability research

Concise portfolio summary: Built a two-LLM writer/checker system that showed AI output can become more accurate through constrained critique and iterative correction, while also revealing why LLM self-correction still requires careful validation design.